United International University

Dept. of Computer Science and Engineering (CSE)

CT 1	Spring 2026
CSE 3812	AI Laboratory
Mark - 15	Time: 30 min

1. Write a Python program that takes an integer input from the user and prints the following: 3

   - the number
   - its square
   - whether it is even or odd

Sample Input	**Sample Output**
7	Number: 7 Square: 49 Odd

2. Write a Python program that takes a number **n** as input and prints the sum of all numbers from **1** to **n** using a **for** loop. 2

Sample Input	**Sample Output**
5	Sum = 15

3. A university computer lab charges students based on the number of hours they use lab computers. 4

   Write a Python program that takes the number of hours as input and calculates the total cost using the following rules:

   - First 2 hours: 50 taka per hour
   - Next 3 hours: 30 taka per hour
   - Remaining hours: 20 taka per hour

   Print the total cost.

Sample Input	Sample Output
2	Total cost = 100
4	Total cost = 160
7	Total cost = 230

4. A student wants to analyze the marks of 5 subjects. 6

   Write a Python program that:

   - takes marks of 5 subjects as input
   - stores them in a list
   - prints the full list
   - using **list slicing**, create the following:
     - a list of first 3 subjects
     - a list of last 2 subjects
   - prints both sliced lists
   - prints the highest mark
   - prints the lowest mark
   - prints how many subjects have marks greater than 80

Sample Input	Sample Output
85, 70, 90, 60, 88	Marks: [85, 70, 90, 60, 88] First 3 subjects: [85, 70, 90] Last 2 subjects: [60, 88] Highest mark: 90 Lowest mark: 60 Count greater than 80: 3